

LECTURE 3: IMPACT OF MODERN DRIVE TRAIN ON ENERGY SUPPLIES

Modern electric and hybrid electric vehicles are becoming an important part of the transportation system. Unlike conventional internal combustion engine vehicles, electric vehicles obtain their energy from electrical sources, mainly through charging from the power grid. Therefore, as the penetration of electric vehicles increases, their charging demand can have a significant effect on the electrical power system. The impact depends on the number of EVs connected to the grid, charging power, charging pattern, location of charging stations, duration of charging, and whether charging is controlled or uncontrolled. At a low penetration level, the existing grid can generally accommodate EV charging, but large-scale and uncontrolled charging may create several problems such as voltage instability, harmonics, voltage sag, increased power losses, transformer overloading, and degradation of power quality.

FUNDAMENTAL EV-GRID LOAD DYNAMICS

The integration of Electric Vehicles (EVs) into the transportation sector marks a paradigm shift in utility load management. From a power systems perspective, an EV acts as a significant mobile load that directly bridges the gap between transportation and the electrical grid.

- **Grid Dependency Analysis:** Unlike traditional internal combustion engine vehicles, EVs do not produce their own primary energy. They function as a grid-dependent load, drawing electricity that must be generated and distributed in real-time.
- **The Load Progression Chain:** The systemic impact follows a deterministic sequence: More EVs → Increased electricity demand → Greater cumulative load on the power system.
- **Comparative Context:** This shift essentially replaces liquid fuel demand (petrol/diesel) with localized and systemic electrical demand, placing the entire energy burden of the transportation sector on the grid infrastructure.

ENERGY GENERATION AND SUPPLY REQUIREMENTS

The large-scale adoption of EV drive trains requires a massive expansion of generation capacity to maintain system balance.

- **Quantitative Generation Study:** To contextualize the scale of additional demand, the following scenario illustrates the energy requirements for a moderate fleet size:

Suppose: There are 1 million EVs. If each vehicle consumes an average of 10 kWh per day, the total additional electricity demand is: $1,000,000 \times 10 = 10,000,000 \text{ kWh/day}$ (10GWH/day). The power generation fleet must be capable of producing this additional 10 GWh daily.

- **Environmental Impact Paradox:** Simply meeting this demand is insufficient for decarbonization. If the grid relies on burning fossil fuels in thermal plants to feed EVs, the pollution levels will not decrease; they will merely shift from tailpipes to smokestacks.
- **Positive Supply Impacts:** If managed correctly, the transition facilitates:
 - Enhanced utilization and integration of renewable energy.
 - Substantial reduction in oil imports.
 - Reduction in greenhouse gas emissions (contingent on a clean energy mix).

TRANSMISSION AND DISTRIBUTION (T&D) INFRASTRUCTURE PRESSURES

The delivery of energy from the plant to the vehicle charger exerts significant stress on the physical components of the grid.

- **The Power Delivery Path:** Electricity must flow through a hierarchical network:
Power Plant → Transmission Lines → Substations → Distribution Transformers → Homes and Charging Stations.
- **Infrastructure Challenges:** T&D networks face four critical hurdles:

1. Requirement for higher current carrying capacity.
 2. Increased line loading.
 3. Heightened transmission and distribution losses.
 4. The necessity for larger, more robust substations.
- **Distribution Overloading Case Study:** Consider 100 homes where each resident installs a 7 kW Level 2 charger. This creates a localized peak demand of 700 kW. If these homes are served by a standard 500 kVA transformer, the transformer will be severely overloaded beyond its thermal and nameplate limits.
 - **Systemic Consequences:** Failure to upgrade infrastructure results in transformer overheating, severe voltage drops, frequent fuse operations, and eventual equipment failure.

PEAK LOAD DYNAMICS AND CHARGING BEHAVIOR

Grid reliability is highly sensitive to the temporal distribution of charging loads.

- **Temporal Demand Variability:** Utility loads typically follow a predictable cycle: Morning (Moderate), Afternoon (Medium), and Evening (High).
- **The 7 PM Peak Scenario:** The primary risk occurs when commuters return home simultaneously at approximately 7 PM and initiate charging. This causes a sharp spike in the existing evening peak, leading to voltage fluctuations, higher generation costs (due to peaking plants), and reduced grid reliability.
- **Load Mitigation Solutions:** Management strategies are essential to flatten the curve:
 - Shifting charging to night or designated off-peak hours.
 - Utilizing automated smart charging systems to stagger load entry.

SMART CHARGING AND V2X TECHNOLOGIES

Modern grid modernization specialists utilize EVs as flexible assets rather than static loads through advanced control logic.

- **Smart Charging Defined:** The intelligent control of charging time and rate based on real-time electricity demand and grid conditions.
- **Economic Incentives:** Utilities use time-of-use pricing to influence behavior. For example, if electricity costs ₹10/unit during the day but only ₹5/unit at night, a smart charger will automatically wait until 11 PM to start the cycle, reducing consumer cost and grid stress.
- **The V2X Ecosystem (V2G, V2H, V2V):**
 - **Vehicle-to-Grid (V2G):** A 60 kWh battery can supply 10 kWh back to the grid during peak demand, recharging later during off-peak hours to provide grid support and emergency backup.
 - **Vehicle-to-Home (V2H) & Vehicle-to-Vehicle (V2V):** These technologies allow the EV to power a household or another vehicle, increasing local resiliency.
- **Virtual Power Plants (VPP):** Groups of EVs acting in aggregate as a single, massive battery. This system acts as a dynamic source or storage system, though it must be noted that frequent cycling can decrease overall battery life.

IMPACT OF MODERN DRIVE TRAIN ON ENERGY SUPPLIES

1. Voltage Instability

Voltage stability is the ability of a power system to maintain acceptable voltage levels under normal operating conditions and following disturbances. Large-scale EV charging can affect voltage stability because EV chargers introduce additional electrical loads into the distribution network. When a large number of EVs charge simultaneously, the current flowing through distribution lines and transformers increases. Because every practical distribution line has impedance, an increase in current produces a voltage drop according to the relationship

$$\Delta V = IZ$$

where I is the current flowing through the line and Z is the impedance of the line. Therefore, when EV charging causes the current to increase, the voltage at the receiving end of the feeder can decrease. In a weak distribution network, long feeder, or heavily loaded system, this voltage reduction can become significant and may result in voltage instability.

EV chargers can also produce rapidly changing power demand because charging power may change according to the battery state of charge and charging-control strategy. Charging and discharging of large numbers of EV batteries can therefore introduce variations in active and reactive power flow. If these variations are not properly controlled, they can contribute to voltage fluctuations and instability. Severe voltage instability can ultimately lead to voltage collapse and, in extreme cases, contribute to widespread power-system disturbances.

Causes of Instability

- **Non-linear Load Characteristics:** Unlike domestic or industrial loads, which are predictable and linear, EV charging draws massive amounts of power in short periods.
- **Unpredictable Power Consumption:** Charging often occurs without coordination, potentially coinciding with "peak hours" when the grid is already under stress.
- **Battery Oscillations:** The physical process of charging and discharging batteries creates damping oscillations that contribute to instability.

Voltage instability caused by EV charging can be reduced by using **smart charging, coordinated charging, voltage regulators, transformer tap changers, reactive-power control, distributed generation, and energy-storage systems**. A smart grid can continuously monitor the grid condition and adjust the charging power of EVs according to the available network capacity.

Real-Time Monitoring Solutions

To prevent grid failure, utilities utilize sophisticated monitoring systems:

A Wide Area Monitoring System (WAMS) is a system used to monitor the electrical power network over a large geographical area. It uses measurement devices such as

Phasor Measurement Units (PMUs) to obtain synchronized measurements of voltage, current, frequency, and phase angle at different locations in the power system.

The basic structure of a wide-area monitoring system can be represented as:

CT/VT → PMU → Phasor Data Concentrator → Monitoring and Control

The **Current Transformer (CT)** measures current, while the **Voltage Transformer (VT)** measures voltage. The PMU processes these measurements and provides synchronized phasor information. Measurements from multiple PMUs are collected by a **Phasor Data Concentrator (PDC)**. The collected information can then be used for visualization, analysis, disturbance detection, system control, protection, and decision-making.

WAMS is particularly useful in an EV-dominated grid because EV charging loads can vary rapidly and may be distributed over a large geographical area. Wide-area measurements allow the grid operator to observe voltage, current, frequency, power flow, and phase-angle conditions at different locations. If abnormal conditions are detected, appropriate control actions can be taken, such as reducing EV charging power, changing transformer tap settings, activating energy storage, or modifying power flows.

2. Harmonics and Power Quality Distortion

Harmonics are sinusoidal components whose frequencies are integer multiples of the fundamental frequency of the electrical system. In a 50-Hz power system, for example, the third harmonic has a frequency of 150 Hz, the fifth harmonic has a frequency of 250 Hz, and the seventh harmonic has a frequency of 350 Hz. In an ideal electrical system, voltage and current waveforms are sinusoidal. However, the presence of harmonics distorts these waveforms.

The main reason EVs can introduce harmonics into the grid is the use of **power electronic converters** in EV chargers. An EV charger generally converts AC power from the grid into DC power required by the battery. This conversion is performed using semiconductor switches and rectifiers. Such converters can behave as nonlinear loads and may draw a nonsinusoidal current from the supply. Consequently, harmonic currents are injected into the distribution system. When these harmonic currents flow

through the impedance of the network, they can also produce harmonic voltage components, resulting in distortion of the supply voltage.

The level of waveform distortion is commonly measured using **Total Harmonic Distortion (THD)**. For current, THD can be expressed as

$$THD_I = \frac{\sqrt{I_2^2 + I_3^2 + I_4^2 + \dots}}{I_1} \times 100\%$$

where I_1 is the fundamental current and $I_2, I_3, I_4, \dots$ are the harmonic currents. A high THD indicates that the current waveform is significantly distorted from its ideal sinusoidal shape.

Harmonics can produce several undesirable effects in the power system. They can increase losses and heating in cables, transformers, motors, and other electrical equipment. They can also reduce the life of insulation, cause malfunction of sensitive equipment and protection systems, and create resonance problems with capacitor banks. Therefore, high EV penetration without proper harmonic control can lead to deterioration of power quality.

Harmonics can be reduced using **power-factor-correction circuits, passive filters, active power filters, improved charger designs, controlled rectifiers, and suitable switching or modulation techniques**. Modern EV chargers are increasingly designed to draw current that is as close as possible to a sinusoidal waveform.

Key Concepts

- **Total Harmonic Distortion (THD):** A mathematical measure of the level of harmonic components in a signal relative to the fundamental frequency.
- **Impact on Equipment:** Harmonics increase stress on power system equipment. This can lead to the burning or bursting of cables and a general degradation of power quality.

Technical Mitigation

- **Harmonic Cancellation:** When thousands of EVs are charged simultaneously, their diverse load patterns, phases, and magnitudes can naturally cancel each other out, provided the chargers are designed differently.
- **Pulse Width Modulation (PWM):** Using PWM in chargers helps eliminate harmonic distortion at the source.

- **Filter Components:** Utilizing Insulated-Gate Bipolar Transistors (IGBTs), active filters, and transistors to minimize harmonic interference.

3. Voltage Sags and Charger Power Limits

Voltage sag is a temporary reduction in the RMS value of supply voltage. It generally lasts for a short duration, typically from about half a cycle to one minute. In a 50-Hz system, one cycle corresponds to 20 ms, so half a cycle corresponds to approximately 10 ms.

EV charging can contribute to voltage sag when a large amount of charging power is demanded suddenly. This is particularly important for high-power DC fast-charging stations. When several high-power chargers start operating simultaneously, a large current is drawn from the distribution network. The increased current flowing through the impedance of the feeder causes a voltage drop according to

$$\Delta V = IZ.$$

As a result, the voltage available to other consumers connected to the same network may temporarily decrease. Voltage sag can affect sensitive electrical and electronic equipment, industrial drives, computers, control systems, and other loads.

The effect can be minimized by using **coordinated charging**, in which the charging of different EVs is scheduled so that they do not all demand maximum power simultaneously. Voltage regulators and transformer tap-changing mechanisms can also be used to maintain the required voltage. Energy-storage systems and local renewable-energy generation can further support the grid during periods of high charging demand.

EV Grid Penetration Thresholds

Research indicates that the impact on the grid is directly tied to the percentage of EVs connected:

- **20% Penetration:** Without management, this level of EV adoption can cause the grid to exceed acceptable voltage sag limits.
- **60% Penetration:** This high level of adoption is only possible without negative impacts if **controlled charging** techniques are used.

Charger Levels and Impact

- **Level 1 Chargers:** Operate at lower voltages (110V). They draw power slowly, causing less voltage sag but requiring significantly more time to charge.

- **Level 2 Chargers:** Fast chargers that draw higher voltage. While they charge batteries quickly, they pose a much higher risk of causing significant voltage sags.

Mitigation

A **smart grid** uses communication, measurement, automation, and control technologies to operate the electrical network more efficiently and reliably. Smart charging is one of the most important techniques for integrating large numbers of EVs into the grid.

In uncontrolled charging, an EV starts charging as soon as it is connected to the charger. In smart charging, the charging time and charging power are controlled according to factors such as grid loading, electricity price, renewable-energy availability, transformer capacity, and the required departure time of the vehicle.

For example, during peak demand, the charging power may be reduced or charging may be temporarily delayed. During off-peak periods, when the grid has sufficient capacity, the EV can be charged at a higher power. This reduces the peak demand and prevents unnecessary stress on the distribution network.

Thus

Smart charging → Peak-load reduction + Better voltage profile + Reduced losses

Power Losses and Transformer Overloading

The penetration of EV charging stations can also increase the power losses in the electrical network. Whenever current flows through the resistance of a conductor, electrical power is dissipated as heat. The resistive power loss is given by

$$P_{\text{loss}} = I^2 R$$

where I is the current and R is the resistance.

When EV charging increases the current flowing through distribution feeders, cables, and transformers, the associated $I^2 R$ losses also increase. The important point is that the loss increases with the **square of the current**. For example, if the current is doubled, the resistive loss becomes approximately four times larger, assuming the resistance remains constant.

Uncoordinated charging can therefore result in greater power losses because many EVs may charge simultaneously. Long-distance transmission or distribution of electricity to

concentrated charging stations can also increase losses. These losses not only waste electrical energy but can also increase the thermal loading of network components.

Power losses can be reduced by using **coordinated charging, local generation, rooftop solar PV, appropriately located charging stations, energy storage, and proper network planning**. Charging EVs using locally generated renewable energy can reduce the amount of power that needs to flow through the distribution network.

Factors Contributing to Loss and Overloading

Uncoordinated Charging: Randomly plugging in vehicles during peak demand periods increases power losses across the system.

Transmission Distance: Power losses increase the further electricity must travel from the station to the vehicle.

- **Transformer Aging:** The heat generated by heavy EV loads accelerates the "rate of aging" for transformers, increasing the risk of failure or fire.

- **Geographical Factors:** In hot regions (e.g., Rajasthan), transformers overheat much faster under EV loads than in cold regions (e.g., Kashmir).

Remediation Strategies

- **Coordinated Charging:** Scheduling charging during "low-load" hours (e.g., after 11:00 PM).

- **Peak Shifting:** Avoiding charging during a city's specific peak hours (such as 4:00 PM in Delhi).

- **Local Generation:** Using rooftop solar panels to charge vehicles locally, reducing the distance power must travel and minimizing transmission losses.

Advanced Grid Integration and The Path Forward

Despite the technical challenges, the transition to modern drive trains is an environmental necessity. IC engines contribute significantly to air pollution and rely on limited oil resources.

Advanced Concepts in Grid Interaction

Modern technology allows EVs to act as more than just consumers of energy; they can become dynamic sources or storage systems:

- **Vehicle-to-Grid (V2G):** EVs return power to the grid during peak demand, assisting the power station.
- **Vehicle-to-Home (V2H):** Using an EV battery to power a house during outages or peak times.
- **Vehicle-to-Vehicle (V2V):** One EV providing a charge to another EV
- **Virtual Power Plants (VPPs):** A concept where a group of interconnected EVs acts as a single, massive battery pack to provide power to the grid

5. Overloading of the Transformer

Distribution transformers are particularly important because they supply electricity to residential, commercial, and EV charging loads. When a large number of EVs are connected to a transformer simultaneously, the transformer must supply additional current. If the total load exceeds the transformer's rated capacity for a significant period, the transformer can become overloaded.

Transformer heating is strongly related to current because copper losses are approximately

$$P_{cu} = I^2 R.$$

Therefore, an increase in EV charging current increases copper losses and consequently increases the temperature of the transformer. Higher operating temperature accelerates the ageing of transformer insulation. Continuous or frequent overloading can therefore reduce the useful service life of the transformer and may eventually cause failure.

The impact of EVs on transformer life depends on several factors, including the number of EVs, charging power, existing transformer loading, charging duration, ambient temperature, and charging pattern. A transformer that already operates close to its rated capacity is much more sensitive to additional EV charging than a lightly loaded transformer.

Transformer overloading can be controlled through **smart charging, coordinated charging, appropriate charging-power limits, transformer capacity enhancement, energy storage, and distributed renewable generation.** Instead of allowing all EVs to charge at maximum power simultaneously, the charging system can limit or shift charging according to transformer loading.

6. Power-Quality Degradation

Power quality refers to the quality and reliability of the voltage and current supplied to electrical equipment. Important power-quality parameters include voltage magnitude, voltage waveform, frequency, harmonics, voltage unbalance, and power factor.

Large-scale EV penetration can affect power quality because EV chargers are power-electronic loads and because simultaneous charging can produce large changes in electrical demand. The combination of harmonic currents, voltage fluctuations, voltage sag, increased current, and phase unbalance can degrade the quality of electrical power supplied to consumers.

For example, if a large number of single-phase EV chargers are connected disproportionately to one phase of a three-phase distribution system, the current loading of the three phases becomes unequal. This produces **voltage unbalance**. Proper distribution of single-phase chargers among the phases and intelligent charging control can reduce this problem.

Therefore, massive EV penetration should be accompanied by appropriate grid planning and power-quality management.

RENEWABLE ENERGY FOR EV CHARGING

The environmental benefit of EVs depends partly on how the electricity used for charging is generated. If additional EV electricity is generated primarily from fossil fuels, emissions may be shifted from vehicle exhaust systems to power plants. If EVs are charged using renewable sources such as solar and wind, the overall environmental benefit can be greater.

Solar PV is particularly suitable for EV charging because charging stations can be installed with rooftop or ground-mounted solar systems. Electricity generated by the PV system can directly supply the charging load or can be supplied through the grid.

Therefore, an integrated system consisting of:

Solar PV + Energy Storage + EV Charging + Smart Grid

can provide a more sustainable approach to EV charging.